Test 4 – Grade 9

Name: _____
Grade / Class: 9

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"Ahmed volunteered at a local animal shelter during his summer break. He helped feed animals, clean cages, and assist visitors in learning about animal care. By participating in this project, he realized the importance of empathy and responsibility. Ahmed's experience inspired him to start a campaign promoting animal welfare in his neighborhood."

a) What activities did Ahmed perform at the shelter?
b) How did the experience change Ahmed's perspective?
c) Suggest two ways the community can help animals.
d) Summarize the passage in your own words in four sentences.

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2. Grammar & Vocabulary

a) Combine using an appropriate conjunction:
"The museum was crowded. We still enjoyed exploring the exhibits."

b) Identify adjectives, adverbs, and nouns in:
"The energetic children played loudly in the colorful playground."

c) Correct the error:
"Neither the teacher nor the students was ready for the surprise quiz."

d) Convert to indirect speech:
"She said, 'I will submit the report by Monday.'"

e) Replace the underlined word with a synonym:
"The book's content was fascinating."

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3. Writing

Write a short essay (120–150 words) on:
“The Importance of Volunteering in Society”
Include benefits, challenges, and your personal opinion.

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4. Literature Analysis

a) Identify the main conflict in a novel you have studied.
b) Explain the use of symbolism in the line:
"The empty chair at the table spoke louder than words."
c) Describe how a character’s choices influence the story’s outcome.

Section B: Mathematics

1. Algebra

a) Solve for x:
3x+7=2x+153x + 7 = 2x + 153x+7=2x+15

b) Factorize:
x2−11x+28x^2 - 11x + 28x2−11x+28

c) Simplify:
6m2n−4mn2+2m2n6m^2n - 4mn^2 + 2m^2n6m2n−4mn2+2m2n

d) Solve simultaneously:
3x+2y=163x + 2y = 163x+2y=16, x−y=3x - y = 3x−y=3

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2. Geometry & Mensuration

a) Find the area of a parallelogram with base 14 cm and height 9 cm.
b) A cone has radius 6 cm and height 10 cm. Find the volume.
c) Distance between points (-3, 7) and (4, -1).

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3. Trigonometry

a) Find cos θ if sin θ = 0.8 and θ is acute.
b) Verify identity: 1+tan2x=sec2x1 + tan^2x = sec^2x1+tan2x=sec2x
c) A flagpole casts a shadow 15 m long. Angle of elevation of top = 45°. Find height of pole.

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4. Ratio, Proportion & Percentage

a) A pair of shoes costs $150. Sold at 10% profit. Find selling price.
b) If 6 liters of juice cost $18, find the cost of 25 liters.
c) Population of a city increased from 250,000 to 275,000. Find percentage increase.

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5. Statistics & Probability

a) Test scores of 10 students: 20, 18, 19, 17, 20, 16, 18, 19, 17, 20. Find mean, median, mode.
b) Construct a frequency table for the above data.
c) A dice is rolled. Probability of getting a prime number.

Section C: Science

1. Biology

a) Describe human respiratory system.
b) Explain the differences between aerobic and anaerobic respiration.
c) Two differences between amphibians and reptiles.
d) Define ecological niche with an example.
e) Explain nitrogen cycle briefly.

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2. Physics

a) A car accelerates from 10 m/s to 30 m/s in 5 s. Calculate acceleration.
b) Define work and give one example.
c) Explain conduction, convection, and radiation with examples.
d) Calculate force if mass = 12 kg and acceleration = 4 m/s^2.
e) State and explain Newton's second law of motion.

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3. Chemistry

a) Identify element, compound, or mixture: Carbon dioxide, Sugar, Air.
b) Write balanced equation for reaction: Zinc + Hydrochloric acid → Zinc chloride + Hydrogen.
c) Difference between ionic and covalent bonds.
d) Define base with one example.
e) Law of conservation of mass – explain with example.

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4. Environmental Science

a) Name two greenhouse gases and their effects.
b) Suggest three ways to reduce plastic pollution.
c) Define desertification and mention two consequences.
d) Explain role of wetlands in ecosystem protection.

Section D: Social Studies

1. History

a) Causes and outcomes of the French Revolution.
b) Significance of Industrial Revolution.
c) Major Axis Powers during World War II.

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2. Geography

a) Define continental drift theory.
b) Name largest mountain range in North America.
c) Two effects of urbanization on environment.

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3. Civics / Political Science

a) Difference between parliamentary and presidential systems.
b) Two responsibilities of citizens in a democracy.

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4. Economics

a) Explain factors affecting supply.
b) Difference between revenue and profit.
c) Importance of financial literacy for teenagers.

Section E: ICT / Computer Studies

1. Computer Basics

a) Explain difference between volatile and non-volatile memory.
b) Name one output and one storage device.

2. Applications

a) Spreadsheet function to find the lowest value in a dataset.
b) Two benefits of digital presentations in classrooms.

3. Programming Basics

a) Define array and give example in Python.
b) Python code to calculate factorial of 6.

4. Cyber Awareness

a) Define phishing and suggest one method to prevent it.
b) Two ways to secure passwords online.

Section F: Islamiyat

1. Quran & Hadith

a) Name two qualities promoted by the Quran.
b) Narrate a Hadith that emphasizes honesty.

2. Aqidah / Belief

a) List the six articles of faith.
b) Explain the concept of Akhirah (life after death).

3. Fiqh / Worship

a) Explain significance of Zakat in Islam.
b) Describe conditions for fasting to be valid.

4. Islamic Ethics & Morality

a) How does Islam promote charity and helping the needy?
b) Suggest two ways to maintain cleanliness as recommended in Islam.

Section G: Arts & Music

1. Visual Arts

a) Explain chiaroscuro in drawing.
b) Three primary colors and the secondary colors they produce.

2. Music

a) Define tempo and dynamics in music.
b) Example of woodwind and percussion instruments.

Section H: Health & Physical Education

1. Physical Fitness

a) Two exercises for upper body strength.
b) Difference between isotonic and isometric exercises.

2. Health & Hygiene

a) Importance of vaccination for teenagers.
b) Two ways to prevent waterborne diseases.

3. Safety & Environment

a) State one safety rule while using chemicals in labs.
b) One method to prevent electrical hazards at home.